Curso Alimente seu Intestino

Por que Intestino ?

Priscila Antunes

Licenciado para Lays Martins 074.232.516-40

Curso Alimente seu Intestino

Bem-vindos às aulas!

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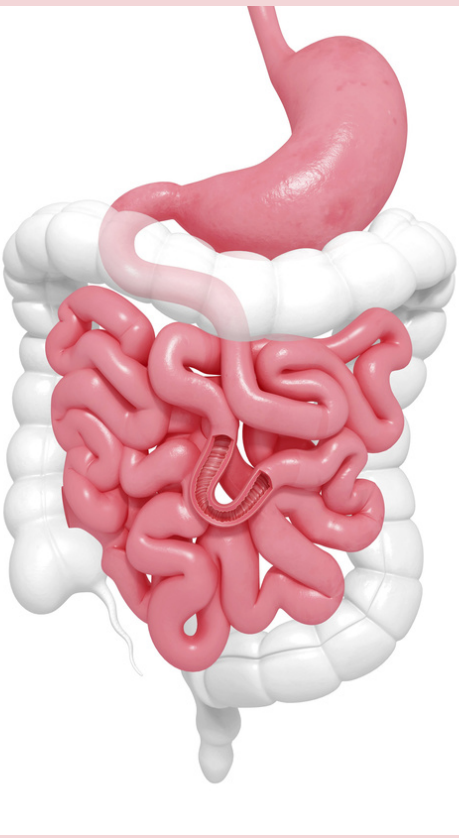
Trato gastrointestinal (boca, estômago, intestino)
Digestão

Introdução à aula

Barreira Intestinal
Microbiota

Sistema Imune

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Processo Digestivo

Digestão - quebra
Absorção - Analítico
Fermentação - transformação
Excreção

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Saliva

Estímulos - odor, sabor, contato, contato, mastigação, náuseas, acidez, etc.

Produção 0,5 a 1,5 L/dia

Ação antibacteriana
Digestão amido



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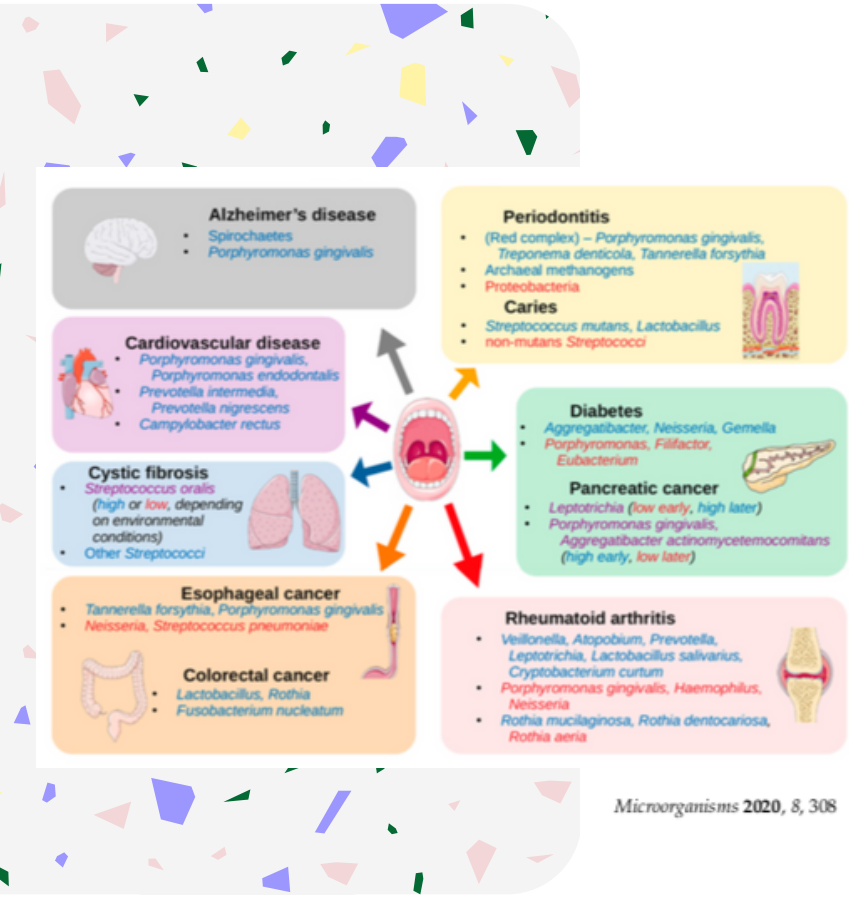
Mastigação





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Microorganisms **2020**, *8*, 308

Microbiota oral

Bactérias em azul estão aumentadas na mucosa oral em pacientes com essas condições, em vermelho estão diminuídas.

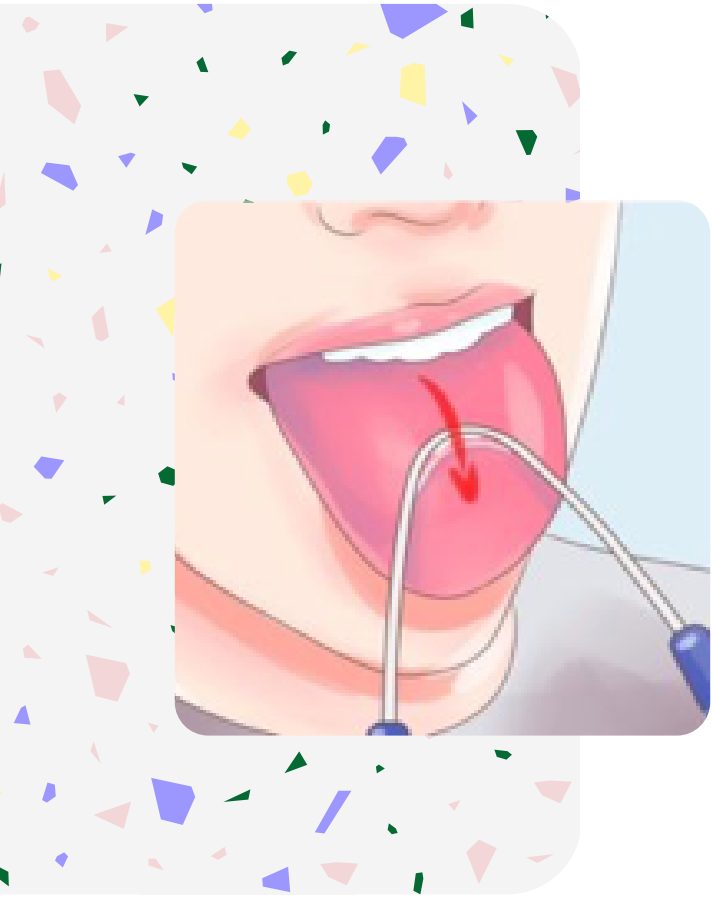
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Ectopic colonization of oral bacteria in the intestine drives TH1 cell induction and inflammation

Intestinal colonization by bacteria of oral origin has been correlated with several negative health outcomes, including inflammatory bowel disease. However, a causal role of oral bacteria ectopically colonizing the intestine remains unclear. Using gnotobiotic techniques, we show that strains of *Klebsiella* spp. isolated from the salivary microbiota are strong inducers of T helper 1 (TH1) cells when they colonize in the gut. These *Klebsiella* strains are resistant to multiple antibiotics, tend to colonize when the intestinal microbiota is dysbiotic, and elicit a severe gut inflammation in the context of a genetically susceptible host. Our findings suggest that the oral cavity may serve as a reservoir for potential intestinal pathobionts that can exacerbate intestinal disease.

Published in final edited form as:
Science. 2017 October 20; 358(6361): 359–365. doi:10.1126/science.aan4526.

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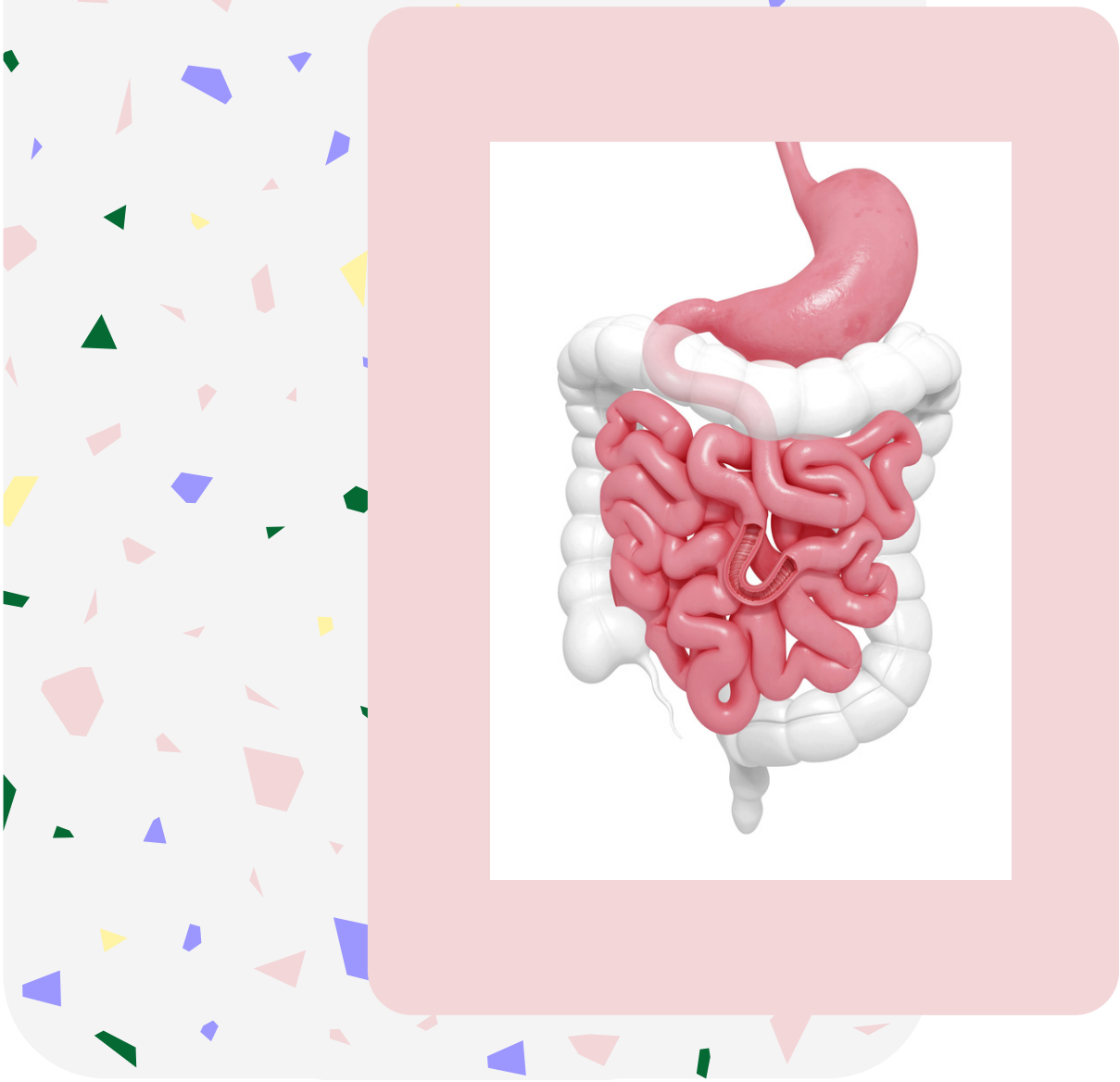
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Microbiota oral

Melhora na digestão, paladar, hálito e aspecto ds fezes.

Complementary Therapies in Medicine, 2018.

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Estômago

Digestão
Produção de ácido clorídrico
Enzimas Digestivas
Camada de muco protetor

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Hipocloridria

- Eructação
- Azia ou refluxo
- Desconforto após comer
- Dificuldade com digestão de refeições protéicas
- Fezes mal cheirosas
- Vitamina B12 baixa
- H. pylori
- Gastrina alta

Hipercloridria

- Azia ou queimação
- Mucosa danificada
- Reação a histamina
- Alergias
- Ansiedade

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Hipocloridria

Enzimas - estômago e pâncreas
Bromelina, papaína, lactase, alfa amilase ...

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Hipercloridria

Verificar histamina
Retirada de alimentos ácidos até a cicatrização
Reparo de mucosa (colágeno, glutamina)
Mucilagens
Chás cicatrizantes
Ciclo circadiano e sono
Retirada de alimentos alergênicos

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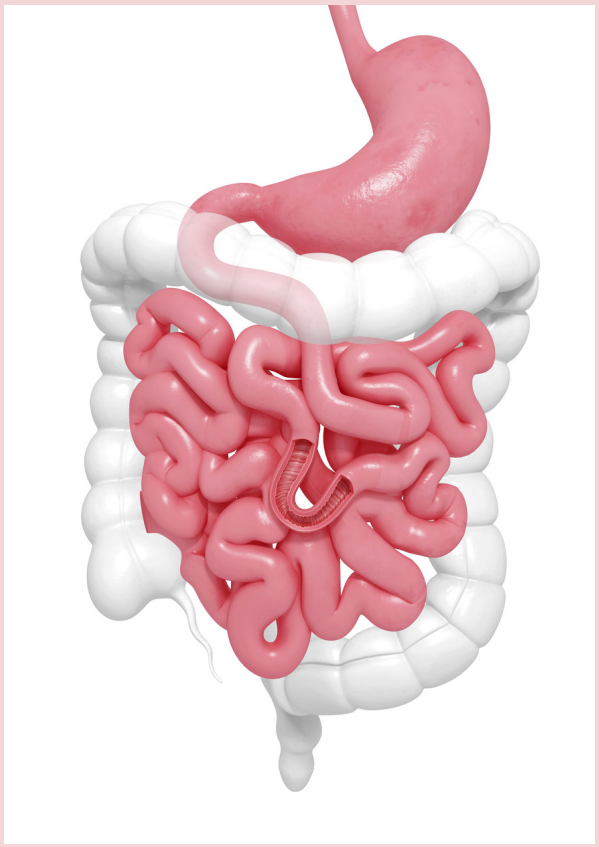
Hipocloridria

Suplementos para proteção de mucosa

Hipercloridria

Suplementos para reparo de barreira intestinal

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Intestino

Absorção
Fermentação
Excreção
Microbiota Intestinal

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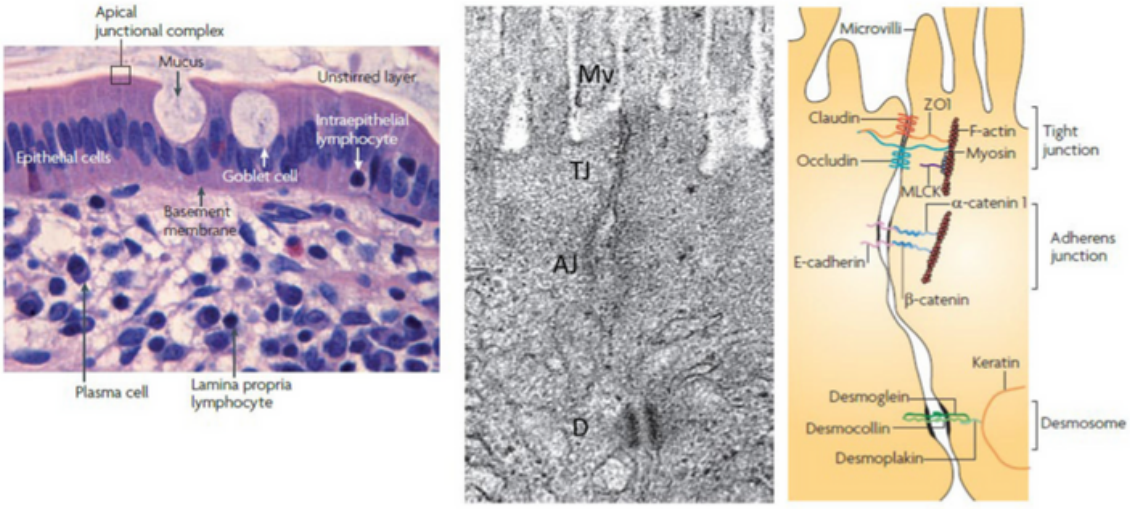
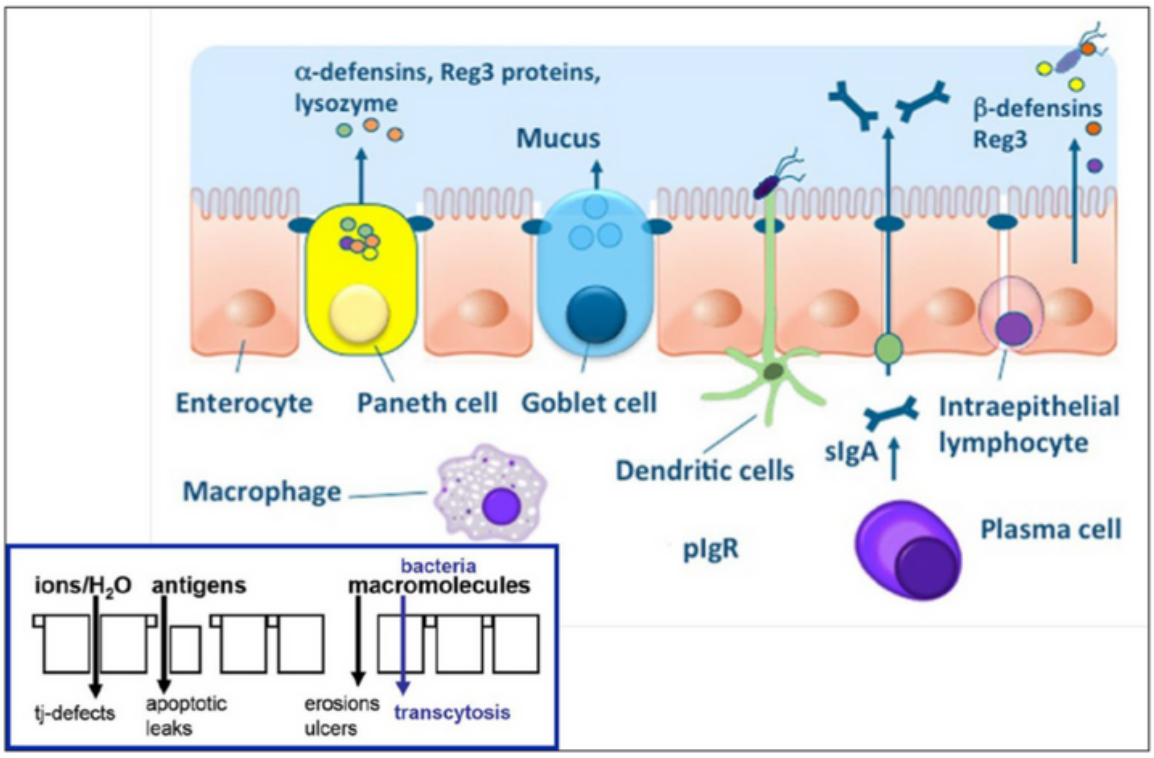


Figure 1: Anatomy of the mucosal barrier.

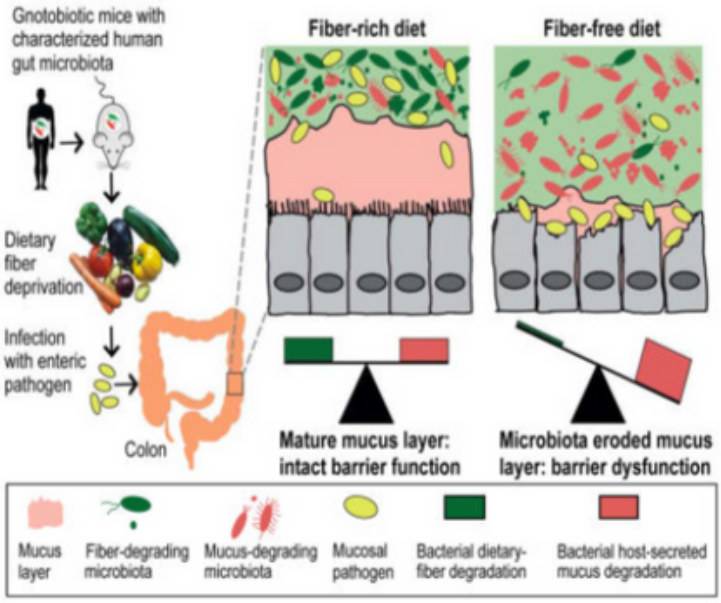
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Gut. 2019 August ; 68(8): 1516–1526. doi:10.1136/gutjnl-2019-318427.

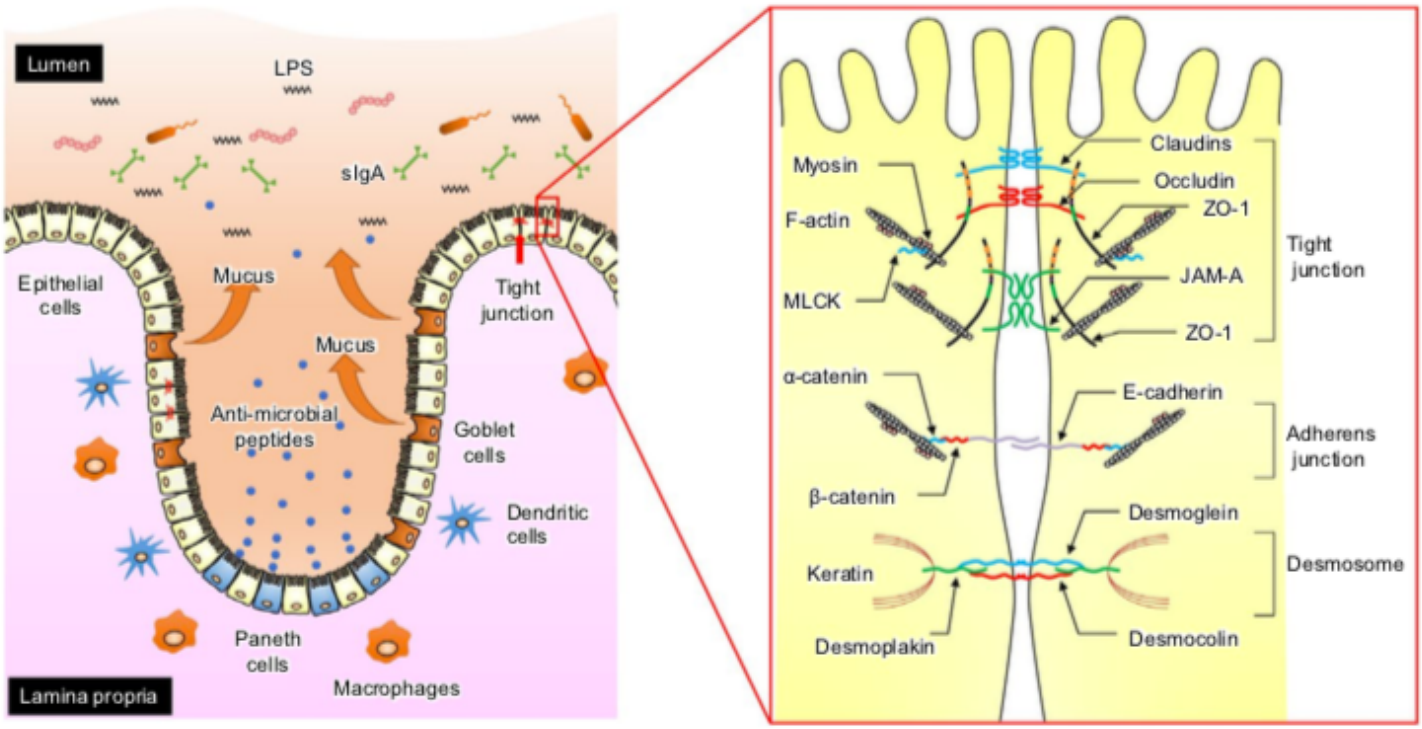
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A dietary fiber-deprived gut microbiota degrades the colonic mucus barrier and enhances pathogen susceptibility



Regular consumption of dietary fiber helps prevent erosion of the intestinal mucus barrier by the gut microbiome, blunting pathogen infection and reducing the incidence of colitis.

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Received: 11 November 2019 | Revised: 29 January 2020 | Accepted: 5 February 2020
DOI: 10.1111/asj.13357

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Crosstalk Between Gut Microbiota and Innate Immunity and Its Implication in Autoimmune Diseases

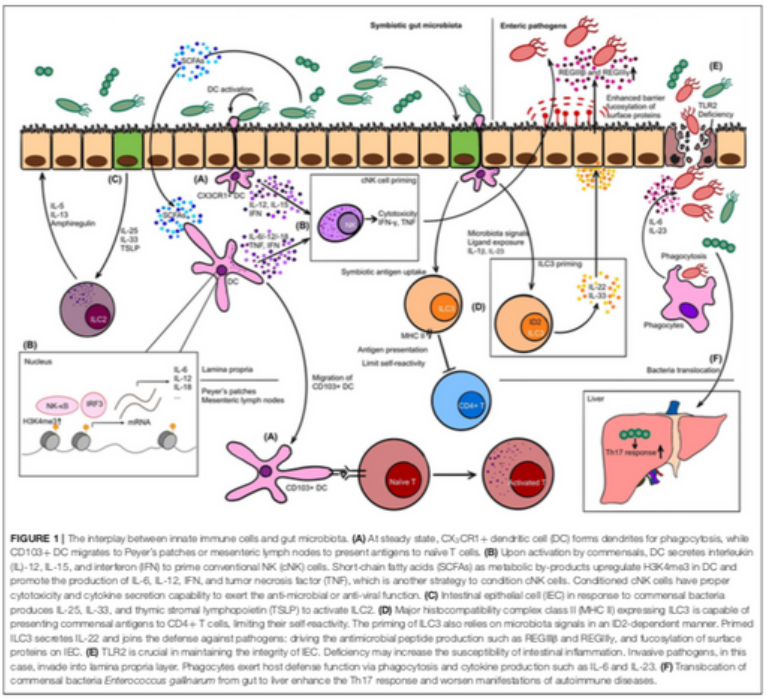


FIGURE 1 | The interplay between innate immune cells and gut microbiota. **(A)** At steady state, CX_3CR1+ dendritic cell (DC) forms dendrites for phagocytosis, while CD103+ DC migrates to Peyer's patches or mesenteric lymph nodes to present antigens to naïve T cells. **(B)** Upon activation by commensals, DC secretes interleukin (IL)-12, IL-15, and interferon (IFN) to prime conventional NK (cNK) cells. Short-chain fatty acids (SCFAs) as metabolic by-products upregulate H3K4me3 in DC and promote the production of IL-6, IL-12, IFN, and tumor necrosis factor (TNF), which is another strategy to condition cNK cells. Conditioned cNK cells have proper cytotoxicity and cytokine secretion capability to exert the anti-microbial or anti-viral function. **(C)** Intestinal epithelial cell (IEC) in response to commensal bacteria produces IL-25, IL-33, and thymic stromal lymphopoietin (TSLP) to activate ILC2. **(D)** Major histocompatibility complex class II (MHC II) expressing ILC3 is capable of presenting commensal antigens to CD4+ T cells, limiting their self-reactivity. The priming of ILC3 also relies on microbiota signals in an ID2-dependent manner. Primed ILC3 secretes IL-22 and joins the defense against pathogens: driving the antimicrobial peptide production such as REGIIIβ and REGIIIγ, and fucosylation of surface proteins on IEC. **(E)** TLR2 is crucial in maintaining the integrity of IEC. Deficiency may increase the susceptibility of intestinal inflammation. Invasive pathogens, in this case, invade into lamina propria layer. Phagocytes exert host defense function via phagocytosis and cytokine production such as IL-6 and IL-23. **(F)** Translocation of commensal bacteria *Enterococcus gallinarum* from gut to liver enhance the Th17 response and worsen manifestations of autoimmune diseases.

REVIEW
published: 21 February 2020
doi: 10.3389/fimmu.2020.00282

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01

Inflamação crônica

02

Cansaço, fadiga, letargia.

Danos na barreira intestinal

Disbiose

03

Sensibilidade alimentar

04

Eixo intestino cérebro: ansiedade, depressão

05

Sistema imunológico

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FLEUMA

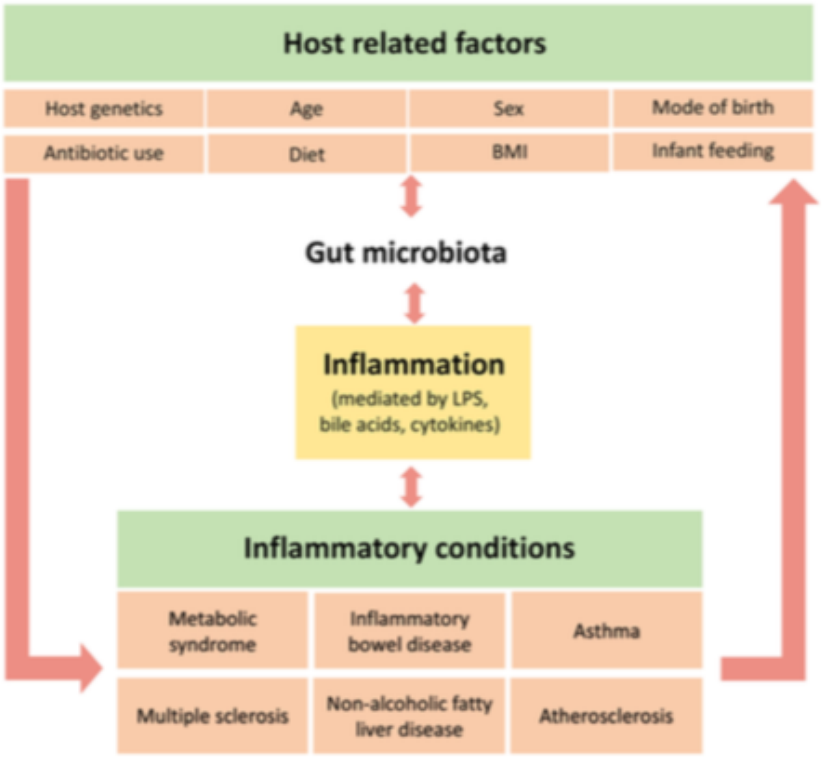
DISBIOSE - INFLAMAÇÃO

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Review

The Gut Microbiota and Inflammation: An Overview

Zahraa Al Bander [1,*], Marloes Dekker Nitert [2], Aya Mousa [1,†] and Negar Naderpoor [1,*,†]



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Tratar a

INFLAMAÇÃO

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Acne



Patients with Acne Vulgaris Have a Distinct Gut Microbiota in Comparison with Healthy Controls

Yongqiong DENG[1#], Hong WANG[1#], Jiyuan ZHOU[3], Yunzhu MOU[4], Guiqiang WANG[2] and Xia XIONG[1]
[1]Department of Dermatology & STD, The Affiliated Hospital of Southwest Medical University, Luzhou, Sichuan, [2]Department of Infectious Disease, Peking University First Hospital, Beijing, [3]Biomedical Research Institute, Shenzhen Peking University, The Hong Kong University of Science and Technology Medical Center, Shenzhen, Guangdong Province, [4]Department of Dermatology & STD, Affiliated Hospital of North Sichuan Medical College, Nanchong, Sichuan, China
#These authors contributed equally to this article.

doi: 10.2340/00015555-2968
Acta Derm Venereol 2018; 98: 783–790

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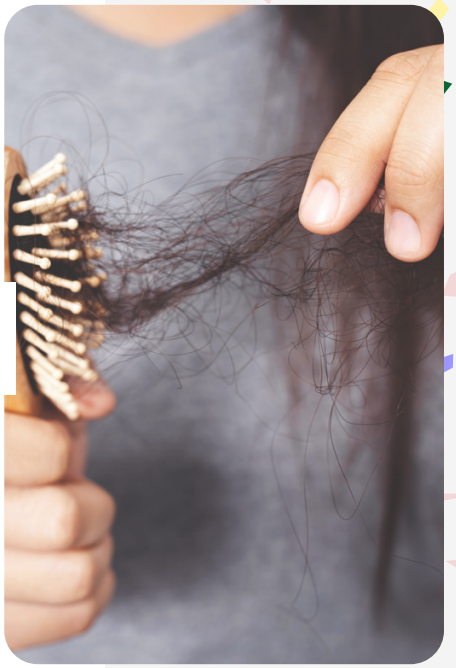
Queda de cabelo

Alopécia

Implication of Human Bacterial Gut Microbiota on Immune-Mediated and Autoimmune Dermatological Diseases and Their Comorbidities: A Narrative Review

Roberta Colucci · Silvia Moretti

Intestinal Dysbiosis and Biotin Deprivation Induce Alopecia through Overgrowth of *Lactobacillus murinus* in Mice



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Alergia



Review

Gut Microbiota as a Target for Preventive and Therapeutic Intervention against Food Allergy

Rosita Aitoro [1], Lorella Paparo [1], Antonio Amoroso [1], Margherita Di Costanzo [1], Linda Cosenza [1], Viviana Granata [1], Carmen Di Scala [1], Rita Nocerino [1], Giovanna Trinchese [1], Mariangela Montella [1], Danilo Ercolini [2,3] and Roberto Berni Canani [1,3,4,5,*]

Nutrients **2017**, 9, 672; doi:10.3390/nu9070672

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Dermatite

Microbiome in the Gut-Skin Axis in Atopic Dermatitis

So-Yeon Lee,[1] Eun Lee,[2] Yoon Mee Park,[3] Soo-Jong Hong[1]*



Review
Allergy Asthma Immunol Res. 2018 July;10(4):354-362.
https://doi.org/10.4168/aair.2018.10.4.354
pISSN 2092-7355 • eISSN 2092-7363

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Psoríase

The gut microbiome in psoriasis and psoriatic arthritis

Bridget Myers [a, 1], Nicholas Brownstone [a, 1], Vidhatha Reddy [a, 1], Stephanie Chan [a, 1], Quinn Thibodeaux [a, 1], Alexa Truong [b, 1], Tina Bhutani [a, 1], Hsin-Wen Chang [a, 2], Wilson Liao [a]

Review

Gut Microbiome in Psoriasis: An Updated Review

Mariusz Sikora [1,*], Albert Stec [2], Magdalena Chrabaszcz [1], Aleksandra Knot [2], Anna Waskiel-Burnat [1], Adriana Rakowska [1], Malgorzata Olszewska [1] and Lidia Rudnicka [1]



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Inflamação

Review

The Gut Microbiota and Inflammation: An Overview

Zahraa Al Bander [1,*], Marloes Dekker Nitert [2], Aya Mousa [1,†] and Negar Naderpoor [1,*,†]

Abstract: The gut microbiota encompasses a diverse community of bacteria that carry out various functions influencing the overall health of the host. These comprise nutrient metabolism, immune system regulation and natural defence against infection. The presence of certain bacteria is associated with inflammatory molecules that may bring about inflammation in various body tissues. Inflammation underlies many chronic multisystem conditions including obesity, atherosclerosis, type 2 diabetes mellitus and inflammatory bowel disease. Inflammation may be triggered by structural components of the bacteria which can result in a cascade of inflammatory pathways involving interleukins and other cytokines. Similarly, by-products of metabolic processes in bacteria,



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Obesidade

Review > Postgrad Med J. 2016 May;92(1087):286-300.
doi: 10.1136/postgradmedj-2015-133285. Epub 2016 Feb 24.

Gut microbiota, obesity and diabetes

Elaine Patterson [1], Paul M Ryan [2], John F Cryan [3], Timothy G Dinan [4], R Paul Ross [5], Gerald F Fitzgerald [6], Catherine Stanton [1]

Review > Nat Rev Gastroenterol Hepatol. 2018 Nov;15(11):671-682.
doi: 10.1038/s41575-018-0025-6.

Gut microbiota–mediated inflammation in obesity: a link with gastrointestinal cancer



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Hipertensão

Colesterol

Review

Gut Microbiota in Hypertension and Atherosclerosis: A Review

Barbara J. H. Verhaar [1,2,*], Andrei Prodan [2], Max Nieuwdorp [2] and Majon Muller [1]



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Coração

Gut Microbiota in Cardiovascular Health and Disease

W.H. Wilson Tang, MD[1,2,3], Takeshi Kitai, MD, PhD[2], and Stanley L Hazen, MD, PhD[1,2]

Review > Transl Res. 2021 Feb;228:109-125. doi: 10.1016/j.trsl.2020.08.007. Epub 2020 Aug 22.

TMAO: how gut microbiota contributes to heart failure



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Fertilidade

Don’t Trust Your Gut: When Gut Microbiota Disrupt Fertility

Mauro S.B. Silva[1] and Paolo Giacobini[1,*]
[1]University of Lille, Inserm, CHU Lille, UMR-S 1172, Laboratory Development and Plasticity of the Neuroendocrine Brain, Jean-Pierre Aubert Research Center, 1 Place de Verdun, 59045 Lille, France
*Correspondence: paolo.giacobini@inserm.fr
https://doi.org/10.1016/j.cmet.2019.09.005



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Menopausa

Influence of Oral and Gut Microbiota in the Health of Menopausal Women

*Angélica T. Vieira[1], Paula M. Castelo[2,3], Daniel A. Ribeiro[3,4] and Caroline M. Ferreira[2,3]**

Review > Maturitas. 2017 Sep;103:45-53. doi: 10.1016/j.maturitas.2017.06.025. Epub 2017 Jun 23.

Estrogen–gut microbiome axis: Physiological and clinical implications

James M Baker [1], Layla Al-Nakkash [2], Melissa M Herbst-Kralovetz [3]



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Parkinson



Brain-gut-microbiota axis in Parkinson's disease

Agata Mulak, Bruno Bonaz

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Alzheimer

Review > J Alzheimers Dis. 2017;58(1):1-15. doi: 10.3233/JAD-161141.

The Gut Microbiota and Alzheimer's Disease

Chunmei Jiang [1], Guangning Li [2], Pengru Huang [1], Zhou Liu [1], Bin Zhao [1]



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Esclerose múltipla

Review > J Neuroimmunol. 2019 Mar 15;328:98-104. doi: 10.1016/j.jneuroim.2019.01.004. Epub 2019 Jan 9.

Gut microbiota in neurodegenerative disorders

Suparna Roy Sarkar [1], Sugato Banerjee [2]

Gut dysbiosis, a primary factor behind various gastrointestinal disorders may also augment lipopolysaccharides, pro-inflammatory cytokines, T helper cells and monocytes causing increased intestinal and BBB permeability via microbiota-gut-brain axis. Consequentially, accumulation of misfolded proteins, axonal damage and neuronal demyelination sets in, thus facilitating the pathogenesis of neurodegenerative disorders like Parkinson's disease, Alzheimer's disease, multiple sclerosis and amyotrophic lateral sclerosis. Studies revealed that intake of probiotics may help in the integrity of intestinal and BBB thus ameliorating the above neurodegenerative disorders. This review summarizes the current understanding of the role of gut microbiota in neurodegenerative disorders and possible intervention strategies.



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Sono

The Role of Microbiome in Insomnia, Circadian Disturbance and Depression

Yuanyuan Li[1], Yanli Hao[2], Fang Fan[1] and Bin Zhang[3]**



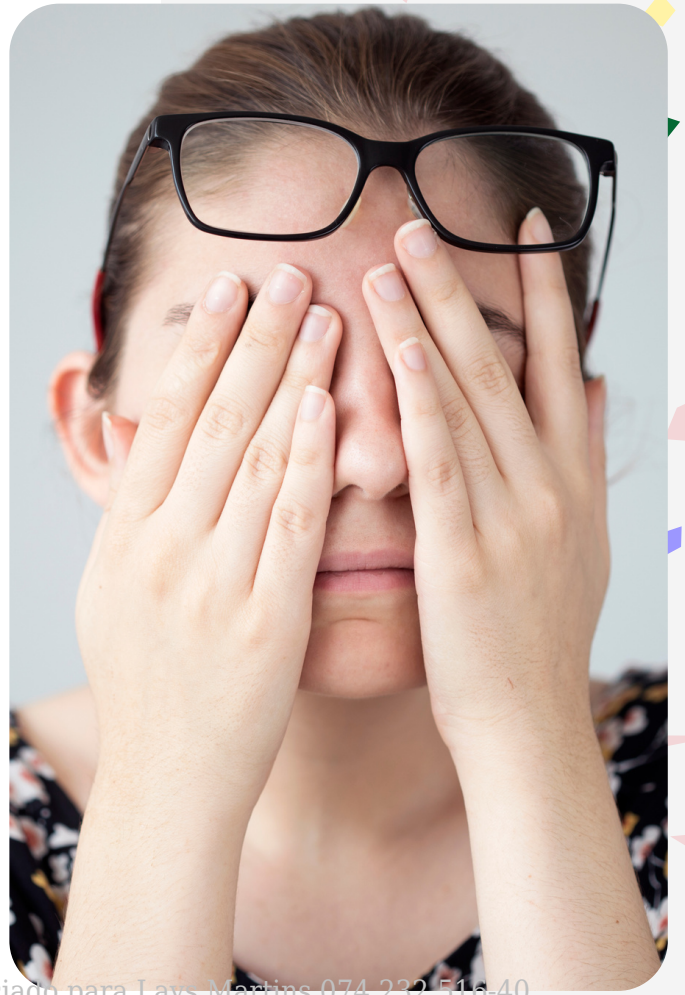
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Fadiga crônica

Review > Mol Psychiatry. 2021 Feb 8. doi: 10.1038/s41380-021-01032-1.
Online ahead of print.

Gut dysbiosis in severe mental illness and chronic fatigue: a novel trans-diagnostic construct? A systematic review and meta-analysis

Jenelle Marcelle Safadi [1] [2], Alice M G Quinton [2], Belinda R Lennox [2], Philip W J Burnet [2], Amedeo Minichino [3]



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Fibromialgia

Gut microbiome: pertinence in fibromyalgia

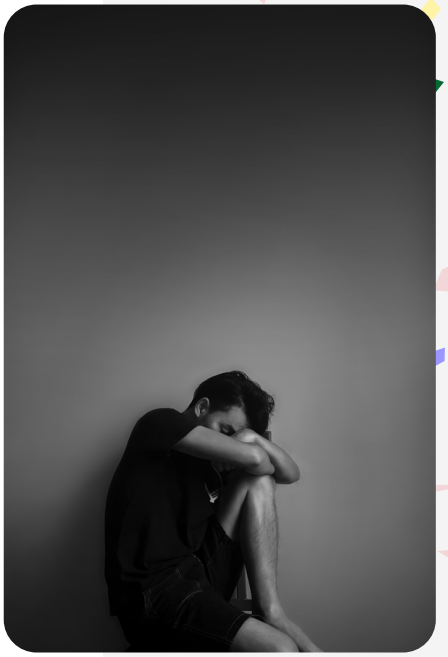
A. Minerbi[1], M.-A. Fitzcharles[2,3]



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Depressão

Ansiedade



Review

The gut microbiota in anxiety and depression – A systematic review

Carra A. Simpson [a, b], Carmela Diaz-Arteche [b], Djamila Eliby [a, b], Orli S. Schwartz [c], Julian G. Simmons [a, b, 1], Caitlin S.M. Cowan [d, 1]

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Massa muscular

Sarcopenia

Gut Microbiota, Muscle Mass and Function in Aging: A Focus on Physical Frailty and Sarcopenia

Andrea Ticinesi [1,2,*], Antonio Nouvenne [1,2], Nicoletta Cerundolo [1], Pamela Catania [1], Beatrice Prati [1], Claudio Tana [1] and Tiziana Meschi [1,2,3]



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Tireoide

Review > Rev Endocr Metab Disord. 2018 Dec;19(4):293-300.
doi: 10.1007/s11154-018-9467-y.

Gut microbiota and Hashimoto's thyroiditis

Camilla Virili [1 2], Poupak Fallahi [3], Alessandro Antonelli [4], Salvatore Benvenga [5 6], Marco Centanni [7 8]



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Asma



The Microbiome in Asthma

Yvonne J. Huang, MD and **Homer A. Boushey, MD**
Division of Pulmonary, Critical Care, Allergy, & Sleep Medicine, Department of Medicine, University of California, San Francisco

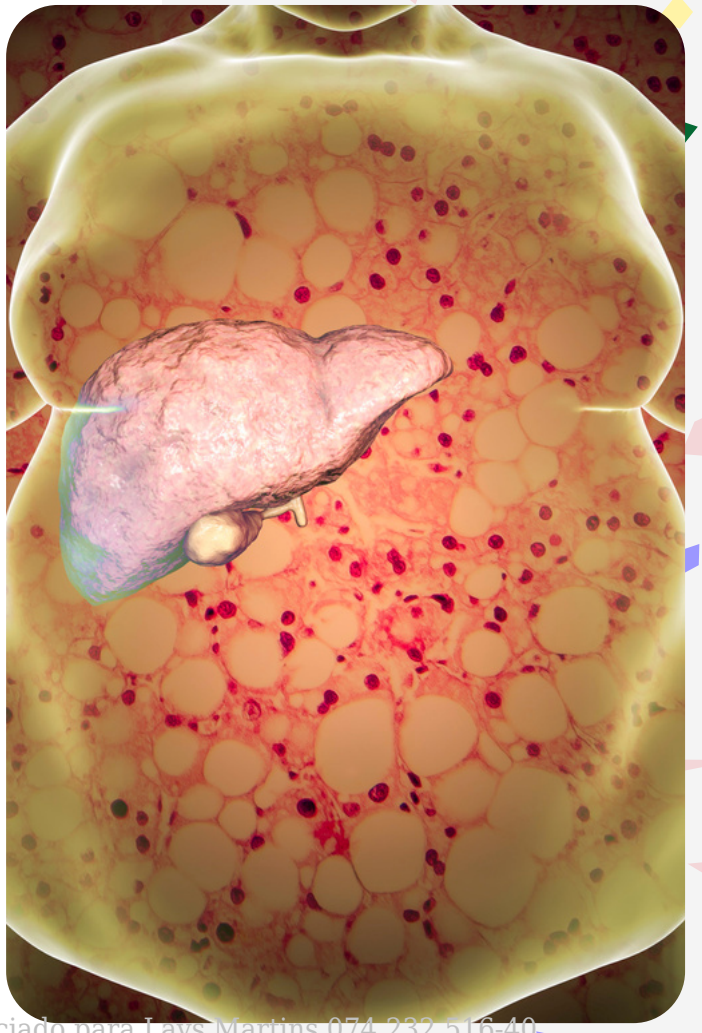
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Fígado

Review > Gut. 2016 Dec;65(12):2035-2044. doi: 10.1136/gutjnl-2016-312729.
Epub 2016 Oct 8.

Gut microbiome and liver diseases

Herbert Tilg [1], Patrice D Cani [2], Emeran A Mayer [3]



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Candida

Review > Virulence. 2017 Apr 3;8(3):352-358. doi: 10.1080/21505594.2016.1247140. Epub 2016 Oct 13.

Fungi in the healthy human gastrointestinal tract

Heather E Hallen-Adams [1], Mallory J Suhr [1]



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SOP

> J Endocrinol Invest. 2021 Aug;44(8):1727-1737. doi: 10.1007/s40618-020-01481-5.
Epub 2021 Jan 2.

Gut microbiota alterations reveal potential gut–brain axis changes in polycystic ovary syndrome

Z Liang [1], N Di [2], L Li [3], D Yang [4]



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Endometriose

> BJOG. 2020 Jan;127(2):239-249. doi: 10.1111/1471-0528.15916. Epub 2019 Sep 19.

Endometriosis and the microbiome: a systematic review

M Leonardi [1], C Hicks [2], F El-Assaad [2], E El-Omar [2], G Condous [1]



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Dor de cabeça

Gut-brain Axis and migraine headache: a comprehensive review

Mahsa Arzani[1†], Soodeh Razeghi Jahromi[2†], Zeinab Ghorbani[3†], Fahimeh Vahabizad[1,4], Paolo Martelletti[5], Amir Ghaemi[6], Simona Sacco[7*], Mansoureh Togha[1,4*] and On behalf of the School of Advanced Studies of the European Headache Federation (EHF-SAS)



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Qual a sua disbiose?

Excesso de bactérias patogênicas
Diminuição de "key species": Akkermansia, Fecalbacterium, prausnitzzii, produtoras de butirato
Diminuição de riqueza e de diversidade
Aumento na proporção de bactérias "boas"

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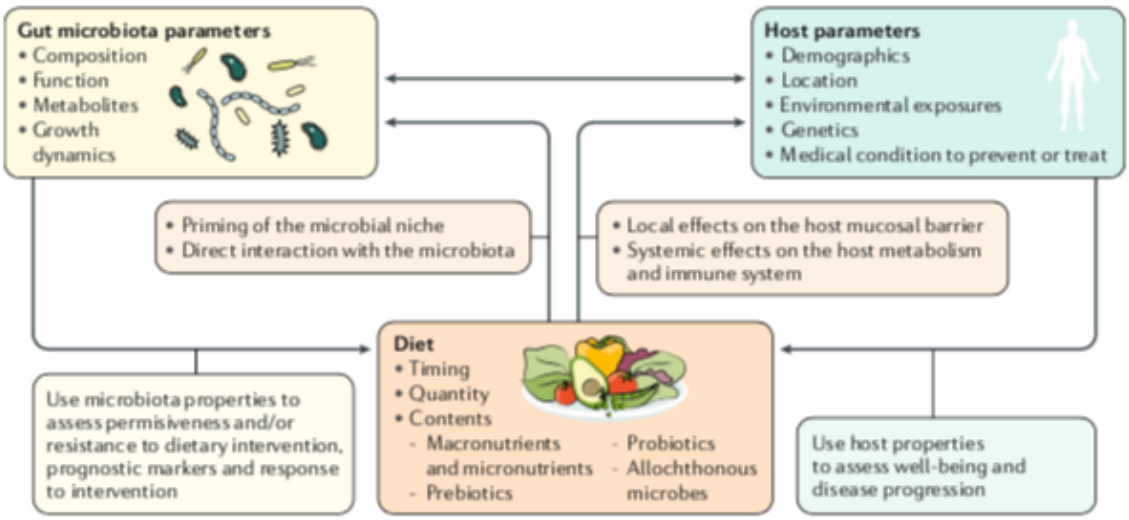
Qual a melhor dieta?



You are what you eat: diet, health and the gut microbiota

Niv Zmora[1,2,3], Jotham Suez[1,3] and Eran Elinav[1]*

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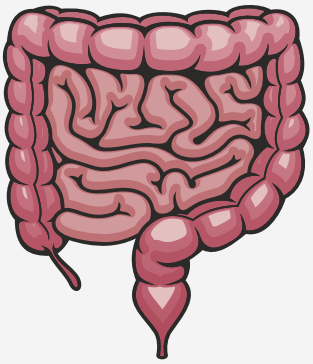


Rotação de estratégias

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Alimente seu Intestino

Priscila Antunes



01

Barreira
Intestinal
suplementação

02

Estresse
RD
Meditação
Atividade
Espiritualidade

03

Alimentação
LFM
Lectina
Alergênicos

04

Microbiota
Prebióticos
Probióticos
Posbióticos
Flavobióticos

05

Digestão
Enzimas
Chás -hortelã,
gengibre,
licorice.

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Resumo

Antes de suplementar, tratar a **DIGESTÃO**
Retirar alimentos pró inflamatórios
Reparar a barreira intestinal
Mudanças com Rotinas Diárias, Meditação, Sono, Natureza
Por fim, probóticos e prebióticos

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